COLUMBIA UNIVERSITY

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Statistical Inference

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Assignment 6

Daniel Rabinowitz

dr105@columbia.edu

Tyler Sotomayor

tyler.sotomayor@columbia.edu

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Tyler Sotomayor
tyler.sotomayor@columbia.edu

Statistical Inference

Instructor: Daniel Rabinowitz

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Maximum Likelihood in Four Models (Problems 1–4)

For each of the following settings, find the following:

- Derive the maximum likelihood estimator for the parameter θ. In the fourth example, define the parameterization by $\theta = \psi'(\xi)$, the derivative of ψ.
- Compute the variance of the estimator.
- Compare the variance of the estimator to the variance bound.

The four settings are those of Assignment 5, and this set finishes what that one started: the two items struck from its instructions are now the instructions. The likelihoods, log likelihoods, and informations below are quoted from last week's write-up rather than rederived. Notation as before: $Y_1, \ldots, Y_n$ the data, $S = \sum_i Y_i$, $\bar{Y} = S/n$; the maximum likelihood estimator is the value $\hat{\theta}$ at which $L(\theta)$, equivalently $\ell(\theta) = \log L(\theta)$, is largest (HPS, §2.4, p. 23). In every setting the variance requested is exact, not asymptotic. That is a small luxury: in general the reciprocal information describes the maximum likelihood estimator only in the large-sample limit (HPS, Property 2, p. 24; DeGroot and Schervish, Thm. 8.8.5, p. 523), and these four models are the unusual case where the finite-sample variance already sits on the bound.

Problem 1

Independent and identically distributed Normal variables with a known variance and unspecified expectation parameter.

Solution:

From Assignment 5, $\ell(\theta) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i (y_i - \theta)^2$ with derivative $\ell'(\theta) = \frac{n}{\sigma^2}(\bar{y} - \theta)$ and information $I_n(\theta) = n/\sigma^2$.

Set the derivative to zero:

$$\frac{n}{\sigma^2}\,(\bar{y} - \hat{\theta}) = 0 \quad \Longrightarrow \quad \boxed{\hat{\theta} \;=\; \bar{Y}} \qquad \text{(likelihood equation)}$$

Since $\ell''(\theta) = -n/\sigma^2 < 0$ everywhere, ℓ is strictly concave and this critical point is the unique maximum. No boundary cases arise; the parameter space is the whole line.

The variance is elementary because the estimator is a sample mean:

$$\mathrm{Var}^{\theta}(\hat{\theta}) \;=\; \mathrm{Var}^{\theta}(\bar{Y}) \;=\; \frac{\sigma^2}{n} \;=\; \boxed{\frac{1}{I_n(\theta)}} \qquad \text{(variance of a mean; Assignment 5)}$$

The bound is attained, at every n and every θ. This is the example HPS uses to introduce efficiency (pp. 17–18), and the attainment is visible without any computation once one notices $\bar{Y} - \theta = \ell'(\theta)/I_n(\theta)$: the estimator's error is a linear function of the derivative of the log likelihood, which is the equality condition in the bound. ■

Problem 2

Independent and identically distributed Poisson variables with an unknown expectation parameter.

SOLUTION:

From Assignment 5, $\ell(\theta) = -n\theta + s\log\theta - \sum_i \log(y_i!)$ on $\theta > 0$, with $\ell'(\theta) = s/\theta - n$ and $I_n(\theta) = n/\theta$.

For $s \geq 1$, the likelihood equation has one root:

$$\frac{s}{\hat{\theta}} - n = 0 \quad \Longrightarrow \quad \boxed{\hat{\theta} \; = \; \bar{Y}} \qquad \text{(likelihood equation)}$$

and $\ell''(\theta) = -s/\theta^2 < 0$, so the root is the maximum. The case $s = 0$ needs its own sentence: then $\ell(\theta) = -n\theta$ is strictly decreasing, the supremum sits at the boundary $\theta \to 0$, and the formula $\hat{\theta} = \bar{y} = 0$ still names the right value, now as a boundary supremum rather than an interior critical point.

For the variance, $\bar{Y}$ is again a mean of independent variables, each with variance θ:

$$\mathrm{Var}^{\theta}(\hat{\theta}) \; = \; \frac{\theta}{n} \; = \; \boxed{\frac{1}{I_n(\theta)}} \qquad \text{(Poisson variance; Assignment 5)}$$

Attained again. DeGroot and Schervish certify this same estimator through the equality condition, writing $\bar{X}_n$ as a linear function of the derivative of the log likelihood (Example 8.8.9, pp. 521–522); here that linearity reads $\bar{Y} - \theta = (\theta/n)\,\ell'(\theta)$. ■

Problem 3

Independent and identically distributed Exponential random variables with unknown expectation parameter.

SOLUTION:

With the mean as the parameter, $f(y \mid \theta) = \theta^{-1}e^{-y/\theta}$ on $y > 0$, and Assignment 5 gives $\ell(\theta) = -n\log\theta - s/\theta$, $\ell'(\theta) = -n/\theta + s/\theta^2$, and $I_n(\theta) = n/\theta^2$.

The likelihood equation:

$$-\frac{n}{\hat\theta} + \frac{s}{\hat\theta^{\,2}} = 0 \quad\Longrightarrow\quad \boxed{\hat\theta \;=\; \bar Y} \qquad \text{(likelihood equation)}$$

Here the second derivative $\ell''(\theta) = n/\theta^2 - 2s/\theta^3$ is not negative everywhere, so concavity has to be checked at the root rather than asserted globally: substituting $s = n\hat\theta$ gives $\ell''(\hat\theta) = n/\hat\theta^2 - 2n/\hat\theta^2 = -n/\hat\theta^2 < 0$, a local maximum; and since $\ell' > 0$ for $\theta < \hat\theta$ and $\ell' < 0$ for $\theta > \hat\theta$ (the sign of $s - n\theta$), the maximum is global. (Exponential variables are positive, so $s > 0$ and the root always exists.)

Each observation has variance θ^2, so

$$\mathrm{Var}^{\theta}(\hat\theta) \;=\; \frac{\theta^2}{n} \;=\; \boxed{\frac{1}{I_n(\theta)}} \qquad \text{(exponential variance; Assignment 5)}$$

Attained. The contrast recorded last week still stands and is worth carrying forward: had the assignment asked for the rate $\beta = 1/\theta$ instead, the maximum likelihood estimator would be $1/\bar Y$, which is biased, and no unbiased estimator of the rate reaches its bound at any finite n (DeGroot and Schervish, Example 8.8.6, p. 520). Asking for the expectation parameter is what makes this problem come out clean. ■

Problem 4

Independent and identically distributed random variables with a density of the form

$$f(y)\exp[\xi y - \psi(\xi)],$$

where f is a specified density, θ is the unknown parameter defined by $\theta = \psi'(\xi)$, and ψ is the function of ξ that makes the expression integrate to 1.

SOLUTION:

Assignment 5 worked this family in its natural parameter: $\ell(\xi) = \sum_i \log f(y_i) + \xi s - n\psi(\xi)$, with $\ell'(\xi) = s - n\psi'(\xi)$, $\mathrm{E}^\xi\{Y\} = \psi'(\xi)$, $\mathrm{Var}^\xi(Y) = \psi''(\xi)$, and information $n\psi''(\xi)$ in ξ. The assignment now names the estimand differently: the parameter is $\theta = \psi'(\xi)$, the mean.

The likelihood equation in ξ is

$$s - n\psi'(\hat{\xi}) = 0, \qquad \text{i.e.} \qquad \psi'(\hat{\xi}) = \bar{y},$$

and $\ell''(\xi) = -n\psi''(\xi) < 0$ (the variance ψ'' is positive for any nondegenerate family), so ℓ is strictly concave in ξ and the root, when it exists, is the unique maximum; it exists whenever $\bar{y}$ falls in the interior of the range of ψ', which is the range of possible means. There is no need to solve for $\hat{\xi}$ itself, because the estimand is $\theta = \psi'(\xi)$ and the likelihood equation already evaluates it:

$$\hat{\theta} \;=\; \psi'(\hat{\xi}) \;=\; \boxed{\bar{Y}} \qquad \text{(likelihood equation, read in the } \theta \text{ parameterization)}$$

This is why the assignment fixes the parameterization before asking for the estimator.

In the natural parameter the likelihood equation says "match $\psi'(\xi)$ to the sample mean"; define θ to be $\psi'(\xi)$ and that instruction becomes the estimator itself.

The variance is exact:

$$\mathrm{Var}(\hat{\theta}) \;=\; \mathrm{Var}(\bar{Y}) \;=\; \frac{\psi''(\xi)}{n}.$$

For the bound, the information must be carried into the θ parameterization. By the reparameterization rule from Assignment 5, with $d\theta/d\xi = \psi''(\xi)$,

$$I_n(\theta) \;=\; I_n(\xi)\left(\frac{d\xi}{d\theta}\right)^2 \;=\; n\psi''(\xi)\cdot\frac{1}{[\psi''(\xi)]^2} \;=\; \frac{n}{\psi''(\xi)},$$

so

$$\mathrm{Var}(\hat{\theta}) \;=\; \frac{\psi''(\xi)}{n} \;=\; \boxed{\frac{1}{I_n(\theta)}} \qquad \text{(reparameterization of the information)}$$

and the bound is attained. (One can reach the same number without reparameterizing: for estimating $m(\xi) = \psi'(\xi)$ the bound in DeGroot and Schervish's general form is $[m'(\xi)]^2/I_n(\xi) = [\psi'']^2/(n\psi'') = \psi''/n$; Thm. 8.8.3, pp. 518–519.)

Problems 1–3 are this problem three times over. The normal with known variance, the Poisson, and the exponential are all of the displayed form after a change of variable, their ψ'' being σ^2, θ, and θ^2 respectively when written as functions of the mean — which reproduces σ^2/n, θ/n, and θ^2/n above. In each case the maximum likelihood estimator of the mean is the sample mean, its variance is exact, and the bound is met. Assignment 5 found the information in the natural parameter, $n\psi''$; this assignment finds it in the mean parameter, n/ψ''; the pair is a worked instance of the rule that information belongs to a parameterization rather than to a model. ■

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